

Toxicity induced by glyphosate and glyphosate-based herbicides in the zebrafish hepatocyte cell line (ZF-L)

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ABSTRACT

Glyphosate is the active component of many commonly used herbicides; it can reach bodies of water through irrigated rice plantations. In the present study, we evaluated the effect of glyphosate and Roundup® (a glyphosate-based herbicide) in established culture of the zebrafish hepatocyte cell line ZF-L after 24 and 48 h of exposure to concentrations of 650 and 3250 µg/L. We observed a reduction in metabolic activity and lysosomal integrity, and an increase in cell number after 24 h of Roundup® exposure at the highest concentration. An increase in active mitochondria and apoptotic cells was observed following 24 h exposure to glyphosate and Roundup®, while only exposure to Roundup® induced an increase in necrotic cells. Rhodamine B accumulation decreased after 48 h exposure to 650 µg/L of Roundup®; this reduction is indicative of increased activity of ABC pumps. Overall, the present findings highlighted the hazard of glyphosate exposure not only in the commercial formulation but also glyphosate alone, since both can induce damage in the ZF-L cell line primarily through the induction of apoptosis.

1. Introduction

Glyphosate-based herbicides are widely used. The mechanism of action of glyphosate is to interrupt the synthesis of essential aromatic amino acids in plants (Giesy et al., 2000). In southern Brazil, this herbicide is used in irrigated rice plantations; therefore, the water used in the crop is drained and returns to the water bodies and thus can cause damage to organisms that inhabit this environment (Anvisa, 2010; Costa, 2008; Primel et al., 2005).

Glyphosate concentrations in rivers around the cultivation are relatively high and may be toxic to organisms that inhabit these areas. Silva et al. (2003) detected a glyphosate concentration of 100 µg/L in a river in southern Brazil 60 days after glyphosate application. Peruzzo et al. (2008) evaluated levels of glyphosate in a river in Argentina and determined a range of 100–700 µg/L.

Several aquatic organisms can survive in contaminated environmental conditions because they have a multixenobiotic resistance system (MXR) (Bard, 2000), which consists of ATP-binding cassette (ABC) proteins, which remove substances from inside the cell and are the primary defense against pollutants (Lehman-McKeeman, 2008). The ABC transporters are a large family of proteins, which includes P-glycoprotein (P-gp), the first identified ABC transporter in teleost fish. P-gp contributes to the efflux of a broad variety of hydrophobic organic

chemicals, neutral or cationic (Sturm and Segner, 2005). Multidrug resistance-associated protein performs the efflux of anionic substances and metabolites of phase II biotransformation (Luckenbach et al., 2014). The biotransformation process is considered a second line of defense and includes the enzyme glutathione S-transferase (GST), which conjugates glutathione to substrates, like xenobiotics, to increase polarity and facilitate excretion, although GST can also act as an antioxidant by combating reactive oxygen species (ROS) (Van der Oost et al., 2003).

Numerous studies have demonstrated the effects of glyphosate-based herbicides related to an alteration in redox balance such as changes in the activity of antioxidant enzymes, lipid damage, genotoxic effects, histological damage, altered acetylcholinesterase activity, and reproductive damage (Cai et al., 2017; Ferreira et al., 2010; Lushchak et al., 2009; Modesto and Martinez, 2010; Sinhorin et al., 2014; Topal et al., 2015; Webster et al., 2014). Reproductive damage has been observed following exposure of both commercial formulations and pure glyphosate in fishes (Harayashiki et al., 2013; Lopes et al., 2014). Guilherme et al. (2014) and Marques et al. (2014) showed a genotoxic effect on *Anguilla anguilla* following exposure to 116 µg/L of Roundup® Ultra. This damage remained 2 weeks after exposure. In vitro experiments also showed a DNA damage after exposure to 42.5 and 85 mg/L pure glyphosate in peripheral blood mononuclear cells and 1.7–4.25 g/

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L in human lymphoma cells (Kwiatkowska et al., 2017; Townsend et al., 2017). Peixoto (2005), using isolated mitochondria demonstrated that both glyphosate and Roundup® were able to perturb the bioenergetic functionality.

When the toxicity of Roundup® and pure glyphosate is compared in aquatic environments, the high toxicity has been attributed to the surfactant polyoxyethylene amine (POEA) (Giesy et al., 2000; Tsui and Chu, 2003). In fact, some studies have demonstrated effects of POEA on the antioxidant defense system, resulting in lipid damage, as Navarro and Martinez (2014) showed in the fish *Prochilodus lineatus* exposed to 0.15–1.5 mg/L of POEA for 24 h. However, the majority of studies performed with pure glyphosate use much higher concentrations than those used in commercial formulations, making comparison difficult.

Studies with comparable concentrations of pure glyphosate and its commercial formulation are necessary to relate the effects of the active component to its commercial formulation. Therefore, the present study aimed to evaluate the toxicity and cellular defense capability in an established culture of zebrafish hepatocytes (ZF-L) following glyphosate exposure and to compare the effects of the active component (glyphosate) with the commercial formulation (Roundup Original®). Several liver cells cultures have been established and applied to toxicity tests of pesticides in an in vitro system, since the liver is a major target organ for most toxins (Huang and Huang, 2012). Also, in vitro studies could characterize the mechanisms by which compounds cause adverse effects besides avoid the animal use, providing results with smaller residues generation, and for this reason ZF-L cell line was chosen to evaluate possible cell targets of this herbicide.

2. Materials and methods

2.1. Cell culture

The zebrafish hepatocyte cell line (ZF-L) was maintained in the Cell Culture Laboratory at the Federal University of Rio Grande at 28 °C in cell culture flasks with 50% L-15 medium and 40% RPMI 1640 medium, supplemented with sodium bicarbonate (0.2 g/L), L-glutamine (0.3 g/L), HEPES (25 mM), 10% fetal bovine serum (Sigma-Aldrich, St. Louis, MO, USA), and 1% antibiotic and antimycotic (penicillin [100 U/mL], streptomycin [100 mg/mL], and amphotericin B [0.25 mg/mL]; Sigma-Aldrich). Phosphate-buffered saline (PBS) enriched with calcium and magnesium (136.9 mM NaCl, 2.7 mM KCl, 1.47 mM KH₂PO₄, 8.1 mM Na₂HPO₄, 1 mM MgCl₂, 0.9 mM of CaCl₂, and pH 7.0) was used to wash cells and photograph cell culture plates.

2.2. Cell exposure

Cells were transferred to 96- or 24-well culture plates (depending on the assay) and incubated for 48 h for complete cell adhesion. Following this, the cells were exposed to glyphosate (analytical standard, Sigma-Aldrich) or Roundup® Original (Monsanto) and analysis was performed after 24 and 48 h.

The solutions of glyphosate and Roundup® were prepared in injection water at a concentration of 200 times higher than final exposure and diluted in culture medium before the distribution to the plate wells. Exposure concentration was 650 and 3250 µg/L for both glyphosate and Roundup® Original (based on the glyphosate present in the formulation). Previous cytotoxicity tests (MTT and NR) were performed in a range of 65–6500 µg/L (data not show) and a reduction in viability began from the 3250 µg/L concentrations in the commercial formulation, chosen as the sublethal concentration in those studies in order to study the pathways of toxicity. The concentration of 650 µg/L was chosen because reflects a concentration which can be found environmentally.

Cell exposure was performed in 96-well plates at a concentration of 1×10^5 cells/mL for the trypan blue, Mitotracker Green, and apoptosis assays, 2.5×10^5 cells/mL for the 3-(4,5-dimethylthiazol-2-yl)-2,5-

diphenyltetrazolium bromide (MTT), neutral red (NR), sulforhodamine B (SRB), and Rhodamine B assays, and 5×10^5 cells/mL in 24-well plates for the GST activity assay.

2.3. Cytotoxicity of ZF-L cells

The cytotoxicity of glyphosate in ZF-L cells was analyzed with several different assays. Metabolism was measured according to Freshney (2005), with the MTT assay. This method consists of the reduction of MTT to formazan crystals through the action of dehydrogenases. These crystals are dissolved in dimethyl sulfoxide and the absorbance is measured at 550 nm with a microplate reader (ELx808, Biotek, Winooski, VT, USA).

Lysosomal integrity was measured with the NR retention assay, which is based on the incorporation and retention of NR by intact lysosomes. If the lysosomes are not intact, the incorporation and retention is diminished, indicating damaged cells (Freshney, 2005). Exposure medium was discarded, and cells were washed twice with PBS, incubated with NR (40 µg/mL) for 3 h, fixed for 5 min in formaldehyde (0.5%) in CaCl₂ solution (1%), and disrupted with acetic acid (1%) in ethyl alcohol (50%). The absorbance was measured at 550 nm with a microplate reader (ELx808, Biotek).

Analyses were performed independently five times with five replicates. Data are expressed as a percentage relative to the control, which was considered 100%.

2.4. Cell quantification following exposure

The number of viable cells was determined by trypan blue exclusion assay, which is based on cell membrane integrity. Viable cells do not retain the trypan blue dye, while non-viable cells exhibit blue staining. Following exposure, the glyphosate-containing medium was discarded, cells were washed twice with PBS, 100 µL of trypan blue (0.04%) was added, and cells were incubated for 10 min at room temperature. The cells were washed twice with PBS and imaged with an epifluorescence inverted microscope (Olympus IX 81). Analysis was performed independently three times with five replicates, with two images taken of each well and counted with ImageJ software. The mean number of cells was used for data analysis.

Cell quantification, based on protein content, was performed as described by Skehan et al. (1990) with modifications described by Gerhardt et al. (2009). After 24 and 48 h of glyphosate exposure, the medium was removed, cells were washed twice with PBS, and fixed for 40 min with 4% formaldehyde (in PBS). Following this, cells were incubated for 1 h with sulforhodamine B (SRB), which binds to basic amino acid residues within proteins, and washed with deionized water to remove unbound stain. The plates were dried at room temperature and proteins were solubilized in 1% sodium dodecyl sulfate. Absorbance was measured with a plate reader at 490 nm. Data are expressed as a percentage relative to the control.

2.5. Active mitochondria

Active mitochondria were measured with the Mitotracker® Green (MTG) assay, according to the manufacturer's instructions (Molecular Probes, Invitrogen, Carlsbad, CA, USA). The MTG probe contains a mildly thiol-reactive chloromethyl that covalently binds mitochondrial proteins, resulting in accumulation in active mitochondria. Analysis was performed independently six times with three replicates. Following exposure, the medium was discarded, cells were washed twice with PBS, and incubated for 45 min with 200 nM of MTG in PBS. This solution was replaced with PBS and images were captured with an epifluorescence inverted microscope (Olympus IX 81). The fluorescence intensity was measured with ImageJ Software and expressed relative to the total number of cells in the area. Data are expressed as a percentage relative to the control group (100%).

2.6. Quantitative analysis of live, apoptotic, and necrotic cells

Cells were evaluated and classified into live, apoptotic, and necrotic according Ribble et al. (2005), through ethidium bromide and acridine orange staining. Analysis was performed independently six times with three replicates. A mixture containing 1 µg/mL of each dye was placed in each well and removed after 1 min to capture the image using an epifluorescence inverted microscope (Olympus IX 81). Three images were captured from each sample and counted with ImageJ Software. Data are expressed as a percentage of live, apoptotic, and necrotic cells. Cells with a green nucleus were considered live, cells with an orange nucleus with fragmented chromatin were considered apoptotic, and cells with a uniformly orange-stained nucleus were considered necrotic.

2.7. Glutathione-S-transferase activity

To evaluate enzyme activity, cells in 24-well culture plates were exposed for 24 and 48 h. After the medium was removed, 0.3% trypsin was added. Three wells were pooled for each sample and centrifuged at $400 \times g$ for 5 min. The pellets were resuspended in phosphate buffer and frozen at -80°C until further analysis. Samples were disrupted ultrasonically in an ice bath for 10 s at 20% (20 kHz; 125 W; 230 V; QSonica Q125, Newtown, CT, USA), and centrifuged at $10,000 \times g$ for 20 min at 4°C . The supernatant was analyzed according to Habig et al. (1974). The change in absorbance was analyzed at 340 nm with an enzyme-linked immunosorbent assay plate reader. Data were expressed as enzymatic activity (nmol/min/mg of protein). Protein quantification was performed with a 3 µL sample on the BioDrop µLITE (England) at 280 nm.

2.8. ATP-binding cassette pump activity

ABC pump activity was indirectly measured with microscopy adaptation according to Smital and Kurelec (1998), with a rhodamine B (ABC protein substrate) accumulation assay. Following exposure, the medium containing glyphosate or Roundup® was removed, cells were incubated with 10 µM of rhodamine B solution for 1 h at 28°C , and the plate was washed twice with PBS. Images were obtained with an epifluorescence inverted microscope (Olympus IX 81). Analysis was performed independently six times with three replicates. The intensity of fluorescence was measured and expressed relative to the number of cells determined with ImageJ Software. Data are expressed as a percentage relative to the control group (100%). The greater the accumulation of rhodamine within the cell, the lower the activity of the ABC pumps.

2.9. Statistical analysis

Each exposure was performed with a minimum of duplicates and repeated at least three times. Data are expressed as a percentage relative to the control group (100%) or mean $\pm$ standard error and were analyzed by analysis of variance following. The prerequisites for homoscedasticity and homogeneity were previously verified. When the differences were significant, the groups were compared by Tukey test. $P < 0.05$ was considered significant.

3. Results and discussion

The results obtained in the present study demonstrated that the effects of glyphosate and glyphosate-based herbicides on ZF-L cells were more pronounced after 24 h of exposure and returned to normal values after 48 h.

We observed a reduction in metabolic activity of ZF-L cells exposed to 3250 µg/L of Roundup® at both 24 and 48 h (Fig. 1A), while lysosomal integrity at the same concentration was reduced only after 24 h of exposure (Fig. 1B). The exposure to glyphosate alone resulted in no

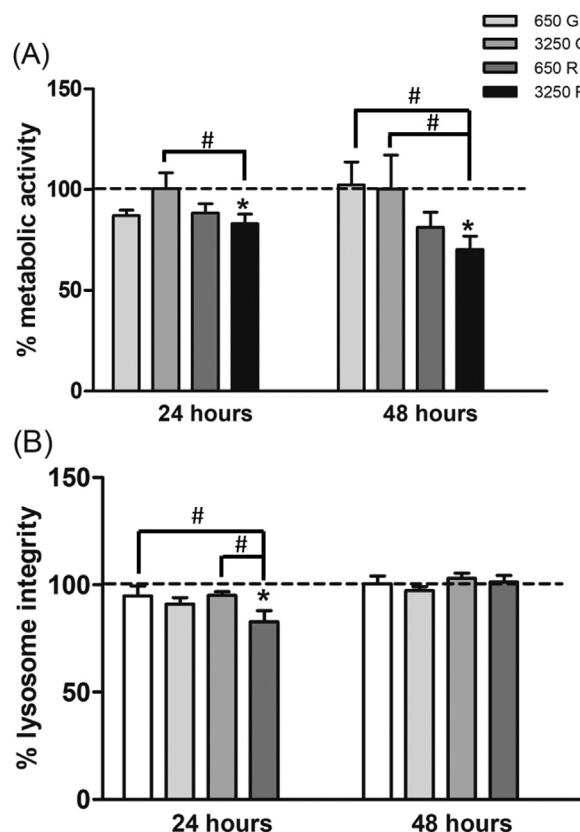


Fig. 1. Cytotoxicity in ZF-L cells determined via the MTT assay (A) and neutral red assay (B) ($n = 5$). Cytotoxicity was measured after 24 and 48 h of exposure to glyphosate (G) and Roundup® (R) (µg/L). Data are expressed as a percentage relative to the control group (dashed line) $\pm$ standard error. Significant differences relative to the control group are indicated by asterisks (*) and differences between groups by pound sign (#).

change in metabolic activity and lysosomal integrity at 24 or 48 h. In the present study, we used the results of the MTT assay to refer to metabolic activity, since MTT reduction, in addition to mitochondrial dehydrogenase activity, can also be attributed to NADH present in the cytoplasm, dehydrogenases associated with the smooth endoplasmic reticulum, and ascorbic acid and sulfhydryl-containing compounds (Stockert et al., 2012).

Both the MTT and NR assays are based on different physiological targets; thus, these assays can demonstrate a different sensitivity in the toxicity pathway of the compounds (Borenfreund et al., 1988). Fotakis and Timbrell (2006) performed a comparison between these assays and indicated the rat hepatoma cell line was more sensitive to the NR assay and the human hepatoma cell line HepG2 was more sensitive to the MTT assay, which indicates that the early sensitivity to compounds may be dependent on cell type or specific cell targets of different compounds. In this study, the MTT assay demonstrated a high sensitivity, showing a reduction in metabolic activity even after 48 h, while the NR assay indicated damage to lysosomal integrity only after 24 h of Roundup® exposure (3250 µg/L).

Cell membrane integrity was evaluated with the trypan blue exclusion assay. There was no significant difference in cell viability (data not shown), while the total number of viable cells increased after 24 h of exposure to the highest concentration of Roundup® and returned to normal values after 48 h (Fig. 2A). However, the quantification of SRB showed no change following glyphosate exposure (Fig. 2B). SRB protein staining has replaced the use of other tests that evaluate cell growth and proliferation and provides improved linearity with the number of cells (Keepers et al., 1991). Nevertheless, the results of our trypan blue assay are not consistent with this. The increase on cell number after 24 h

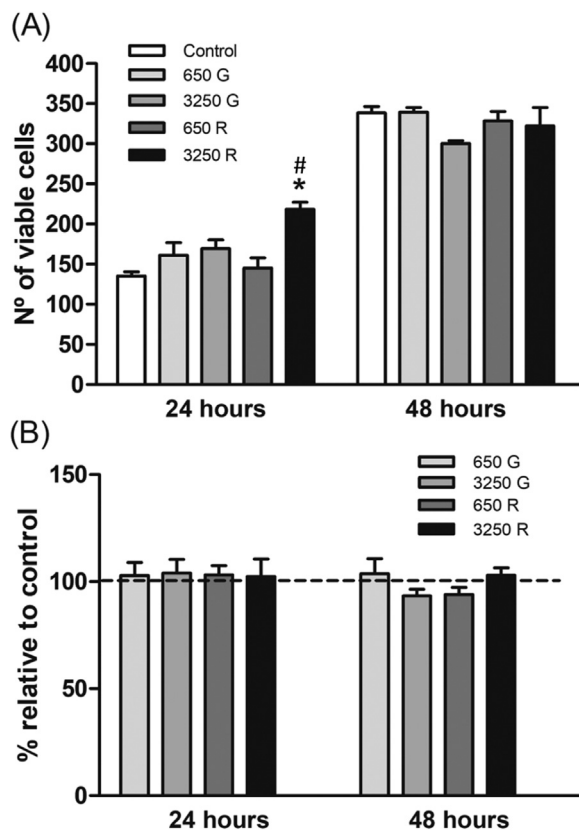


Fig. 2. ZF-L cell quantification by trypan blue exclusion assay (A) and sulforhodamine B assay (B) ($n = 3$ and $n = 4$, respectively). Cell quantification was measured after 24 and 48 h of exposure to glyphosate (G) and Roundup® (R) ($\mu\text{g/L}$). Data for the trypan blue assay are expressed as the mean $\pm$ standard error. Data for the sulforhodamine B assay are expressed as a percentage relative to the control group (dashed line) $\pm$ standard error. Significant differences relative to the control group are indicated by asterisks (*) and differences between groups by pound sign (#).

could be due to the other substances present in the commercial formulation, which can cause the induction of cell proliferation initially and after that the cells grow are stagnate because the reduction on metabolism activity or by induce necrosis (see below).

A reduction in cell membrane and lysosomal integrity were observed by Goulart et al. (2015) after 6 h of ZF-L exposure to Roundup® Transorb at concentrations of 67.7, 135.4, and 270.8 $\mu\text{g/L}$. A reduction in metabolic activity was also observed following exposure at 135.4 and 270.8 $\mu\text{g/L}$. In another study, diploid and triploid fin cells (DIMF and TRMF, respectively) of *Misgurnus anguillicaudatus* exposed to concentrations between 80 and 1040 mg/L of glyphosate showed an LC_{50} of 325.34 and 371.77 mg/L, respectively, and cytotoxicity, as a reduction in metabolic activity, was observed at the lowest tested concentration (Yanjie et al., 2017). Thus, in this study, both pure glyphosate and the commercial formulation exhibited cytotoxic potential, while in the present study no cytotoxicity was observed for pure glyphosate in the conventional assays. This may be due to the concentration used, which was much lower than in the cited study, as well as different sensitivities of the cell lines used.

Our results demonstrated an increase in active mitochondria after 24 h of exposure to the highest concentration of glyphosate and both concentrations of Roundup®, which returned to normal values after 48 h (Fig. 3).

Other studies have reported changes in mitochondrial function following glyphosate or Roundup® exposure. Using isolated rat liver, Peixoto (2005) showed inhibition of some mitochondrial enzymes of the electron transport chain complex following Roundup® exposure, but

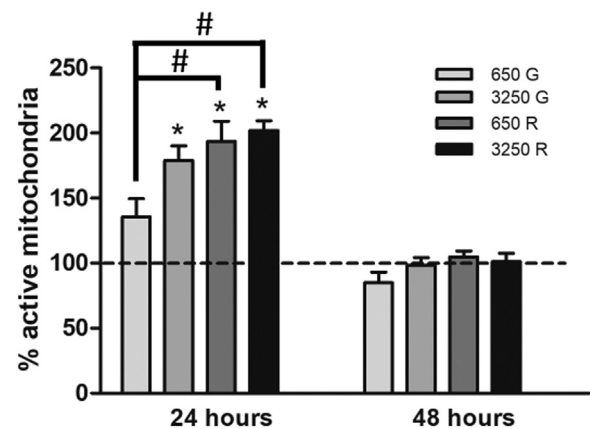


Fig. 3. Quantification of active mitochondria in ZF-L cells using Mitotracker® Green ($n = 6$). Active mitochondria were measured after 24 and 48 h of exposure to glyphosate (G) and Roundup® (R) ($\mu\text{g/L}$). Data are expressed as a percentage relative to the control group (dashed line) $\pm$ standard error. Significant differences relative to the control group are indicated by asterisks (*) and differences between groups by pound sign (#).

not glyphosate alone, at concentrations ranging from 85 to 850 mg/L. A reduction in mitochondrial function was observed in fish sperm following exposure to Roundup® and pure glyphosate. Harayashiki et al. (2013) observed this reduction following exposure of the fish *Poecilia vivipara* to 0.7 mg/L of Roundup® for 96 h and Lopes et al. (2014) demonstrated the same response profile following exposure of the fish *Danio rerio* for 24 and 96 h to 10 mg/L of pure glyphosate.

Given that mitochondria are the energy producers of the cell, our results show that although the percentage of active mitochondria increased, metabolic activity showed a decrease with the highest concentration of Roundup® after 24 h of exposure. Glyphosate may also cause a reduction in metabolic activity at other concentrations, but the large number of active mitochondria may mask the effect. That could mean that the increase in active mitochondria is a compensatory response to low metabolic activity, where in the case of glyphosate alone the observed increase in active mitochondria at 24 h (Fig. 3) was sufficient to maintain cellular metabolism, with no difference compared with the control group (Fig. 1A).

The analysis of apoptosis and necrosis showed a decrease in the number of live cells at 24 h (Fig. 4A) at the highest concentration of glyphosate and at both concentrations of Roundup®. Moreover, all groups exhibited an increase in the number of apoptotic cells when compared with the control group, while only the higher concentration of Roundup® demonstrated an increase in necrotic cells. After 48 h of exposure, these alterations were no longer observed (Fig. 4B).

Many studies have demonstrated the ability of glyphosate based-herbicides to induce apoptosis. The number of apoptotic cells increased in rat pheochromocytoma PC12 cells following 24 h exposure to 6.76-g/L glyphosate, and these values continued to increase after 48 and 72 h (Gui et al., 2012). Martini et al. (2012) exposed 3T3-L1 fibroblasts to 240 $\mu\text{g/L}$ of glyphosate-containing Atanor formulation for 24 h and observed an increased in apoptotic cells and caspase-3 activity.

Kim et al. (2013) showed an increase in apoptosis and necrosis and a decrease in mitochondrial membrane potential following exposure of rat heart cells to 0.85 and 1.7 mg/L glyphosate, but only in combination with the surfactant TN-20. The authors of this study attribute these effects to a change in the toxicodynamics of glyphosate induced by the surfactant. In the present study, we showed an induction of apoptosis and an increase in active mitochondria after 24 h of exposure to both Roundup® and pure glyphosate.

The induction of apoptosis may be due to an alteration of the mitochondrial membrane potential caused by glyphosate exposure, as Astiz et al. (2009) showed in rat livers exposed to 10 mg/kg of

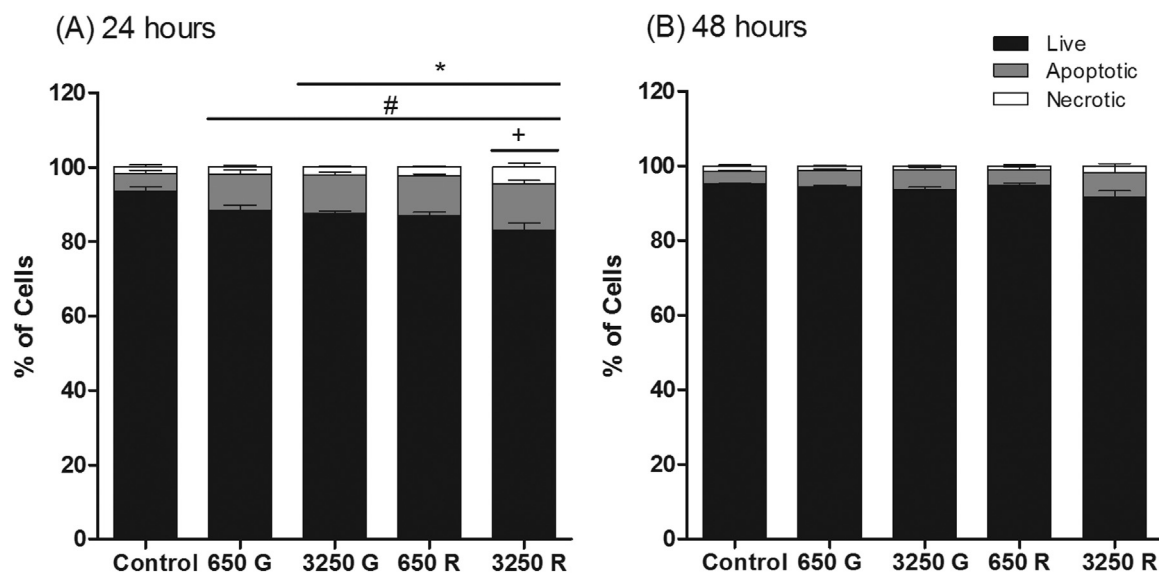


Fig. 4. Induction of ZF-L cell death ($n = 6$). Quantification of apoptotic and necrotic cells after 24 (A) and 48 h (B) of exposure to glyphosate (G) and Roundup® (R) ($\mu\text{g/L}$). Data are expressed as a percentage relative to the control group (dashed line) $\pm$ standard error. Significant differences are indicated by * compared with live cells, by # compared with apoptotic cells, and by + compared with necrotic cells.

glyphosate for 5 weeks. Activation of the caspase cascade can also initiate apoptosis. [Chaufan et al. \(2014\)](#) demonstrated that exposure to glyphosate and glyphosate formulations at a concentration of 3.39 g/L increased caspase-3/7 activation-dependent apoptosis in the human hepatoma cell line HepG2. Another mechanism of apoptosis activation is increased intracellular calcium, which can release cytochrome c and activate the apoptotic pathway. [Cavalli et al. \(2013\)](#) observed an increase in cell death by induction of calcium uptake in rat Sertoli cells in primary culture exposed to glyphosate at 36 $\mu\text{g/L}$ for 30 min.

According to [Meurette et al. \(2005\)](#), a depletion of glutathione (GSH) levels can also induce apoptosis, since GSH plays an important role in protecting against oxidative damage and the production of peroxidative products of DNA and proteins. GST also plays an important role in cell defense, by conjugating GSH with xenobiotics to increase elimination.

In the present study, GST activity was reduced following 24 h exposure of both glyphosate and Roundup® at higher concentrations but only the reduction in Roundup® group remained after 48 h, indicating a recovery on glyphosate group (Fig. 5). [Modesto and Martinez \(2010\)](#) showed a decrease in GST activity follow by a decrease in GSH content

in the fish *Prochilodus lineatus* after 6 and 24 h of exposure to 1 and 5 mg/L of Roundup® Transorb. GST activity returned to normal levels after 96 h while GSH content continued to increase ([Modesto and Martinez, 2010](#)). After 90 days of exposure to 38 mg/L of the glyphosate-based herbicide Nongtoshi, a decrease in GSH level was observed in the plasma of goldfish ([Li et al., 2017](#)). However, other studies, such as [Ferreira et al. \(2010\)](#) and [Moura et al. \(2017\)](#), have demonstrated an increase in GST activity in the liver of fishes exposed to Roundup®. According to [Yanjie et al. \(2017\)](#), DIMF and TRMF cells also exhibited an increase in GST activity following 24 h of glyphosate exposure. Overall, these studies suggest that the effects may depend on the type of exposure (in vivo or in vitro), the cell type, or the type of glyphosate-based formulation.

We hypothesized that this reduction may be due to an inhibition of GST by indirect effects of glyphosate exposure. In fact, [Hermes-Lima and Storey \(1993\)](#) demonstrated in an in vitro assay the inhibition of GST activity by products of oxidative stress, such as hydrogen peroxide. Since the generation of ROS are natural in mitochondria during the production of energy ([Murphy, 2009](#)), the increase observed in its activity could increase the leakage of electrons and consequently the ROS generation will be higher. Other studies showed that glyphosate-based herbicides may be able to increase ROS production in non-target organisms, which was observed by [Sánchez et al. \(2017\)](#) in livers of *Jenynsia multidentata* after 96 h of exposure to 0.5 mg/L of Roundup® Transorb. This was also observed in *Caenorhabditis elegans* exposed to TouchDown, another commercial formulation of glyphosate, at concentrations between 5.5% and 9.8% for 30 min ([Bailey et al., 2017](#)). Similarly, increased production of ROS was observed in zebrafish embryos after 96 h of pure glyphosate exposure at 5, 10, and 100 mg/L.

ABC pumps comprise an important group of membrane transporters that control the movement of compounds out of cells and confer the MXR phenotype ([Luckenbach et al., 2014](#)). In the present study, we observed an increase in MXR activity after 48 h of exposure to Roundup® (650 $\mu\text{g/L}$), which was evident with the smallest accumulation of the substrate Rhodamine B (Fig. 6). This increase may reflect the cell attempting to eliminate glyphosate or the output of metabolic waste. According to [Kurelec et al. \(1998\)](#), the MXR mechanism could be activated or inhibited by many chemical compounds. [Goulart et al. \(2015\)](#) showed an inhibition of this system through an increase in Rhodamine B accumulation following ZF-L exposure to 67.7 $\mu\text{g/L}$ of Roundup® Transorb. The fluorescent substrate Rhodamine B is not

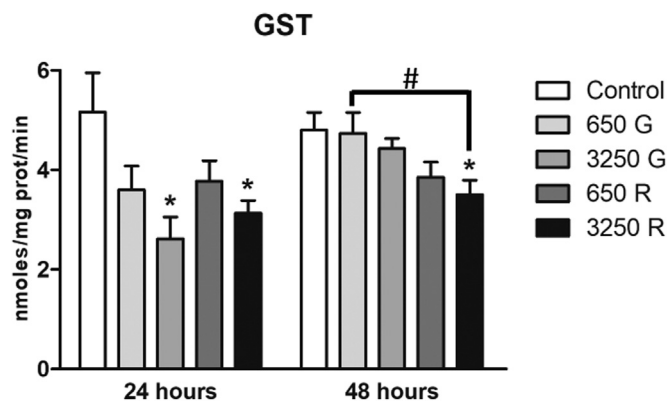


Fig. 5. Glutathione-S-transferase activity in ZF-L cells ($n = 7$). Enzymatic activity was measured after 24 and 48 h of exposure to glyphosate (G) and Roundup® (R) ($\mu\text{g/L}$). Data are expressed as the mean $\pm$ standard error. Significant differences relative to the control group are indicated by asterisks (*) and differences between groups by pound sign (#).

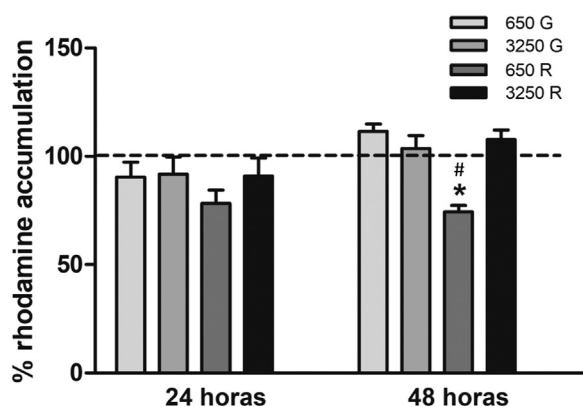


Fig. 6. Activity of ABC pumps through rhodamine B accumulation in ZF-L cells (n = 6). Rhodamine accumulation was measured after 24 and 48 h of exposure to glyphosate (G) and Roundup® (R) (µg/L). Data are expressed as a percentage relative to the control group (dashed line) ± standard error. Significant differences relative to the control group are indicated by asterisks (*) and differences between groups by pound sign (#).

specific for one type of ABC pump; it acts as a substrate for all ABC pumps present in the cell membrane. Thus, given that different families of ABC proteins transport different compounds, one type may be more active than another and/or glyphosate may stimulate one type and inhibit another.

Moreover, ABC pumps are ATP-dependent, and thus their activity may be reduced with a lack of available ATP. Kwiatkowska et al. (2016) observed a reduction in ATP levels in human peripheral blood mononuclear cells exposed to 0.85–1.7 g/L of pure glyphosate for 24 h. In the present study, we did not measure ATP levels but the increase in active mitochondria followed by a reduction or no change in metabolic activity suggests there may have been some damage. This could be related to ABC pump activity, since after 48 h of exposure, when the percentage of active mitochondria returned to the levels of the control group, an increase in ABC pump activity was observed.

Overall, the damaging effects of glyphosate primarily occurred at 24 h of exposure, while after 48 h, ABC pumps are more active and GST activity returns to normal, and the effects apparent at 24 h are no longer observed. It is possible that the compounds have already been biotransformed in the cell or even eliminated in 48 h, and although the increase in pump activity occurred only in the lower concentration of Roundup®, we have the hypothesis that different subfamily are responding to this exposure, which may be resulting in the induction of some subfamily types as well as a reduction in others, showing a similar activity to the control. With respect to apoptosis and necrosis, we can attribute the induction of apoptosis to glyphosate, and in commercial formulations, the damage generated is higher and can lead to cell death due to necrosis, possibly by the joint action of other components of the formula.

According to Giesy et al. (2000), when the toxicity of glyphosate and the Roundup® formulation are compared, the toxicity of aquatic environments has been attributed to the surfactant POEA. The relative toxicity of POEA is higher than Roundup®, which in turn is higher than the toxicity of glyphosate (Tsui and Chu, 2003). Janssens and Stocks (2017) exposed damselfly to glyphosate and Roundup® at concentrations of 1 and 2 mg/L during the larval stage and observed a decrease in the growth rate and scape swimming speed at both concentrations, although the effects were more apparent with the commercial formulation. Zebrafish exposed to 10 mg/L of both glyphosate and Roundup® exhibited increased mortality and reproductive toxicity and an acceleration of the percentage of animals hatched (Webster et al., 2014).

These results are consistent with those of the present study, which indicates that the commercial formulation is more toxic, but the active

component glyphosate is not harmless in the absence of surfactant. In fact, when glyphosate groups present harmful effects to the cell there was no significative difference with the commercial formulation, which make us to suggest that not only the damage effects are due to the surfactant or inert compounds in the formulation but by the glyphosate by itself, that has may have a greater effect in the cell by the greater membrane cross ability conferred by the formulation. Our results showed that glyphosate alone was able to increase the number of apoptotic cells, in addition to increasing the number of active mitochondria and decreasing GST activity, an important cellular defense enzyme.

4. Conclusions

Both glyphosate and Roundup® exposure resulted in ZF-L cell damage, primarily following 24 h of exposure, increasing mitochondrial activity and the percentage of apoptotic cells and decreasing GST activity. These effects were more evident with Roundup® exposure, in which we observed reduction in the MTT and NR analyses. Moreover, damage effects of glyphosate alone were not visible through the classical cytotoxicity tests (MTT and NR), been observed only in the apoptosis test, which suggest that these tests are not sensible to glyphosate exposure despite leading to cell death.

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